

DR Code: The Two Dimensions Barcode supporting high rate data recovery

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Abstract— This study proposed DR Code (Data Restorable Code) which is the two-dimension barcode endurable to the data loss. We aimed to present the design and affirm that our proposed DR Code is easier to be restored than the QR Code being broadly used today.

As H-level QR Code, a model of QR Code that can be restorable the most, in which 30% of the data could be recovered from two types of the data loss, we found that our DR Code could recover 33% of the lost data and applicable for six cases of data loss. Besides, this DR Code applies a simpler algorithm of the data loss in which XOR is used instead of the complicated Reed-Solomon error correction. Thus, our DR Code can be rapidly processed and easier to be implemented, as well as being applicable for industry and logistics sectors.

Keywords—DR Code; QR Code; Barcode; data recovery; Data restorable Code;

I. INTRODUCTION

The barcode system has been broadly known since 1970 which, at that time, was one-dimension barcode that support only the digital data. Later, it was modified to support various types of data e.g., ASCII, Unicode, etc. It also was developed to store more data within limited space and able to recover some data partially lost from the barcode.

Nowadays, the industry and logistics sectors broadly use the two-dimension barcode and place it on the product parcel which could be possibly scratched or damaged. The mostly popular one to be used is the QR Code since users can use the H-level of the QR Code to recover 30% of the lost data which happens on the bordering areas of the barcode. These areas are the circle in the middle and the upper row [1] as identified with the black line in Fig. 1.



Fig.1. The area of QR code allows to damage[1] (black areas)

If the data loss happened to 30% of the bordering areas below as illustrated on the left in Fig. 1, it was impossible to be recovered since the necessary format information of the QR Code, especially, Error Correction Level [2] was damaged too.

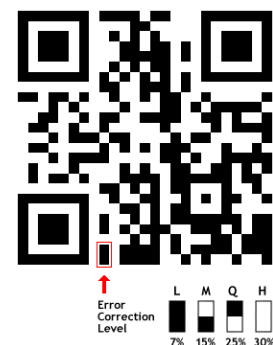


Fig.2. The Error Correction Level stored location of the QR code [2]

However, for the practical use, the lost data would be a horizontal row which could happen as the row 1, 2 and 3 as presented in Fig. 3. Otherwise, if the product parcel was 90° turned upside down, it is possible that the scratching could be as the row 4, 5, and 6.

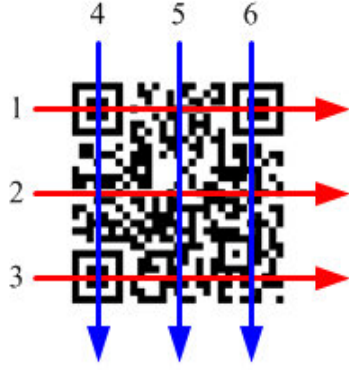


Fig. 3. The damage in a straight line, which often happens.

This study aimed to propose the new model of the two-dimension barcode that supports the damages that could happen to the barcode; although the six cases of data loss happen or the damage happens on its subset area, it is possible for the system to recover all the lost data. In this regard, the research presents the background and related works in Section 2; meanwhile, the Basic Design will be explained in Section 3. Section 4, 5, and 6 will present Improvement Design, Experimentation and Result, as Conclusion, respectively.

II. BACKGROUND AND RELATED WORKS

In 1970, IBM Company created and developed UPC symbol to be used for inputting the data automatically to POS (Point of Sales) system. This system supported 13-digit data. Later in 1974, it was developed to store both digital and character data with 30 digits. This system was called Code 39. Soon after in 1980, Code 49 and Code 16K were proposed and could store the data with 100 digits. The leaping of the barcode system happened in 1994 together with an opening of the QR Code that could store 7,000 digit data and it could record various types of the characters including Kanji characters broadly used in Japan. Therefore, the QR Code is the most popular barcode at the present time. Fig. 4 illustrates the Evolution of the Barcode [1].

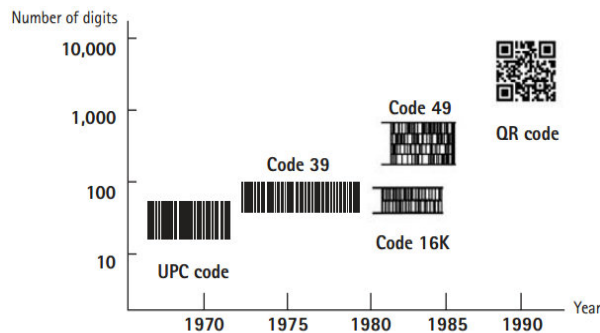


Fig. 4. Evolution of the barcode.

In addition to the QR Code, the two-dimension barcode is currently and gradually being developed. Different types of

barcode serve different purposes; for example, Devi Parikh, Gavin Jancke [3] suggested that using color as an additional factor to enable the barcode to store more data. Later, Bulan et al. [4] proposed dot orientation to enable the barcode reader to easily identify different colors. On the other hand, Yu Fang [5] focused more on designing the two-dimension barcode that suits being read by smart phone. Differently, Tan and Chai [6] studied on reading the two-dimension barcode from the JPEG files.

However, neither research nor studies has been conducted intensively on the design of the two-dimension barcode that could better support the data loss than the QR Code. From the prior study, we found that the zone that the QR Code allows the data loss is the circle area in the middle and the upper row (see Fig. 1). Therefore, this study aimed at developing our DR Code to solve that problem.

III. BASIC DESIGN

Our purpose is to recover the lost data though the loss might be as much as 33%, but the damaged data must be on the determined area which supports six cases of the data loss as presented in Fig. 5 (the dark area). If one third of the barcode was damaged and it happened on the determined area (or in the subset area) of the mentioned six cases, it is still possible to recover all of the lost data.

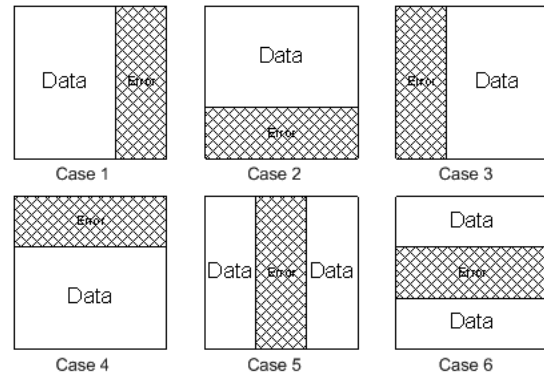


Fig. 5. Various cases of damage.

Additionally, we also applies the simpler algorithm that is easier for the system implementation (programming is easy for all platforms). Thus, the XOR could be defined as follows.

If C is the result of A XOR B,

$$C \text{ XOR } A = B$$

$$C \text{ XOR } B = A$$

For example: when a text (sample data from the QR Code) “ABCDEFGHJKLMNOPQRSTUVWXYZ” was cut into two parts i.e. Line-1 and Line-2. Then, an XOR between these two lines would be resulted as Line-3 (the XOR of string will be byte by byte).

Line-1: ABCDEFGHIJKLM

Line-2: NOPQRSTUVWXYZ

Line-3: \$%#!@*)^\$#<&%

If Line-2 XOR Line-3, it will result as Line-1.

If Line-1 XOR Line-3, it will result as Line-2.

Based on the principle, the barcode data arrangement was designed as shown in Fig. 6 in which all of the data was classified into six blocks including A0, A1, A2, B0, B1, and B2, respectively. These blocks were put into the barcode that has been prepared for the 9-block data. So, the free space in the barcode is reserved for C0, C1, and C2, as " $C_i = A_i \text{ XOR } B_i$ " as follows.

$$C0 = A0 \text{ XOR } B0$$

$$C1 = A1 \text{ XOR } B1$$

$$C2 = A2 \text{ XOR } B2$$

A0	B2	C1
A1	B0	C2
A2	B1	C0

Fig. 6. The information placement in the barcode.

Therefore, if the data loss occurs in six cases as mentioned, it is possible to recover the lost data; for example, in case 2 that the data on the lower row was damaged (see Fig. 7 and 6), it can be calculated to find A2 from $A2 = B2 \text{ XOR } C2$, to find B1 from $B1 = A1 \text{ XOR } C1$, and to find C0 from $C0 = A0 \text{ XOR } B0$, etc.

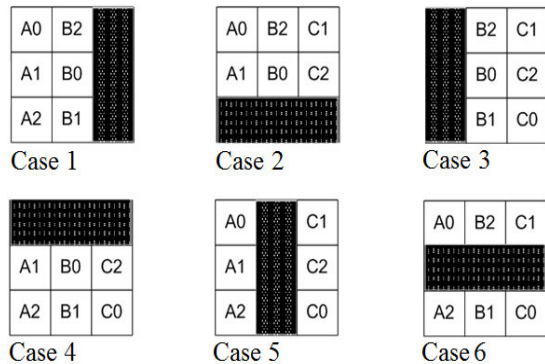


Fig. 7. The barcode information placement in various cases of damage.

IV. IMPROVEMENT DESIGN

We have developed the model in section 3 and made it ready for the practical implementation by adding Finder Pattern into all four corners (see Fig. 8), in order to assist the barcode reader to find the boundary of the barcode more rapidly. The shape of the Finder Pattern is similar to that of the QR Code which adopted and developed from the MaxiCode [10]. We use Direction Signature to remind the barcode reader about the upper row (i.e., A0, B2, and C1), the lower row, the left row, and the right row. This help to find the position of data blocks. Indeed, although some data and Direction Signature might be lost (in any of the six cases), the system can still find the position of a data block.

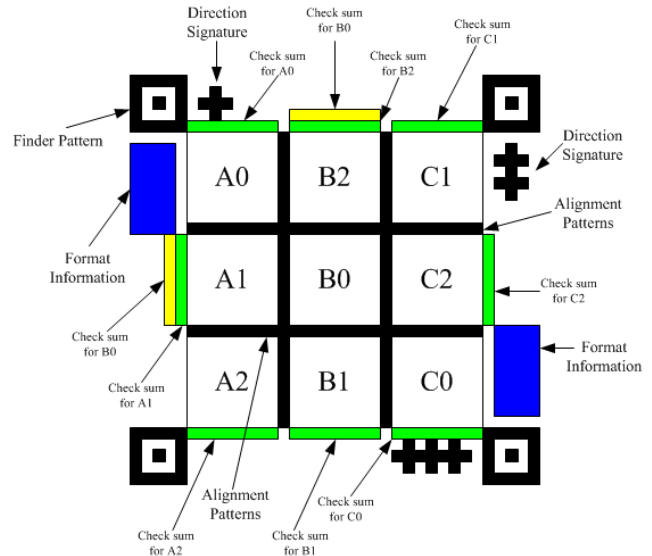


Fig. 8. Improvement of Design

To verify whether the data in each data block contains any errors, we calculated the Check Sum by adding the data from the blocks byte by byte. In each adding calculation, if the result is overflow, the overflowing bit will be dropped out. All bytes of data within the particular block will be added together to be the "Final Number" which is one-byte data. All bits of the Final Number will be converted to be its opposite bit value (i.e., from 10110010 to be 01001101). This converted byte is called "Check Sum" and will be stored into the bar code nearby its data block.

During the process of the barcode reader, to verify if the data block contains errors, it can be calculated by using this Check Sum method except converting the Final Number to be Check Sum. Instead, the Final Number will be added with the existing Check Sum from the bar code. If the result was 11111111 in binary, it is affirmed that no error was found. On the other hand, if the result varies other than 11111111, it is affirmed that it contains some errors.

Since B0 is a data block in the middle and surrounded with Alignment Pattern (with a size of 1 bit for horizontal and vertical rows), it is possible to put its Check Sum next to its data block. Also, in order to help the barcode support the data loss in all six cases, it is necessary to put the Check Sum of B0 on two positions as shown in Fig. 8.

In the data block (e.g. A0) where the data can be placed horizontally, the smallest size of the DR Code can store 8 bytes of data in each data block, in which this 8-byte data will be put byte by byte on 8 separate rows, so in total the smallest DR Code can store 72 bytes of the data. However, to enlarge the data size, it is recommended to expand the wide range of the block byte by byte (8 bits) and the number of the row in the block should match the number of bits (the width of the column). Thus, the bigger DR Code ($N=2$, when N = number of the bytes on a row in the block) will store 32 bytes of data in each block and the total size of data will be 288 bytes as illustrated in Table 1.

TABLE I. RELATIONSHIP BETWEEN N, THE NUMBER OF BYTES IN EACH BLOCK, AND THE TOTAL NUMBER OF BYTES.

N	Bytes per block	Total bytes
1	8	72
2	32	288
3	72	648
4	128	1,152
5	200	1,800
6	288	2,592
7	392	3,528
8	512	4,608
9	648	5,832
10	800	7,200
11	968	8,712
12	1,152	10,368

Hence, the width (or height) of the DR Code can be obtained from: (Finder Pattern Size (5 cells) $\times$ 2) + (Alignment Pattern Size (1 cell) $\times$ 2) + (8 $\times$ N $\times$ 3)

So, the width of the DR Coded is (5 $\times$ 2) + (1 $\times$ 2) (8 $\times$ 1 $\times$ 3) = 36 cells as shown in Fig. 8, and if needed to store the 7,000-byte data, the width should be = 12 + (8 $\times$ 10 $\times$ 3) = 252 cells.

Information about N, actual data size (which has not been added with pad or space to be 72, 288, 648, ...), and a version of the DR Code will be stored in Format Information (See Fig. 8).

V. EXPERIMENTATION AND RESULT

The proposed model was implemented with Java to develop the DR Barcode Generator and DR Barcode Reader on Android Ver. 2.3 on the smart phone, Samsung Galaxy S II. A comparative study was conducted on its operation with the QR Code with the three most popular free QR Code readers [11] e.g. QR Droid, Barcode Scanner, and RedLaser Barcode & QR Scanner, respectively. We also generated the QR Code through web application on [racoindustries.com](http://www.racoindustries.com) [12] (<http://www.racoindustries.com/barcodegenerator/2d/qr-code.aspx>). The implementation was done also with different types of text data and it was repeated for 30 times. The result was analyzed to find the mean to be recorded on the table.

The result from the speed test on the barcode reading process without data recovery was illustrated in Table 2. The result of the data recovery by counting the times of data recovery completed within 10 seconds was shown in Table 3.

TABLE II. RESPONSE TIME

Data Size (bytes)	Response Time (second)			
	DR code	QR Droid	Barcode Scanner	QR Scanner
70	0.246	3.749	2.537	4.264
280	0.271	3.275	2.845	4.370
640	0.317	3.768	2.719	5.265
1,100	0.329	3.590	3.638	4.846
1,800	0.284	5.426	3.527	5.867
2,500	0.342	5.041	3.844	6.365
3,500	0.390	5.375	3.681	6.280
4,600	0.373	5.784	3.562	8.243
5,800	0.521	7.264	4.847	-
7,000	0.595	9.254	4.354	-

Note: For the QR Code with its data recovery tested on the QR Droid, Barcode Scanner, and RedLaser Barcode & QR Scanner, we created the QR Code through web application on [racoindustries.com](http://www.racoindustries.com) by defining the value on the row of QR Code Error Correction Level as "H (30% of code words can be restored)" and simulated the data loss by covering 15 – 20% of the upper area on the QR Code with a white color paper.

From the result in Table 2, it was found that our DR Code reading was completed very quickly since the verification of the Check Sum and the decoding process applies the simpler algorithm than those in the QR Code. When comparing with the QR Code decoding processed by the three mostly used free QR Code readers software, it was found that Barcode Scanner functioned with maximum speed followed by QR Droid, and RedLaser Barcode & QR Scanner, respectively.

TABLE III. THE NUMBER OF SUCCESSFUL DATA RECOVERY WITHIN 10 SECONDS.

Data Size (bytes)	The number of successes in 10 seconds			
	DR code	QR Droid	Barcode Scanner	QR Scanner
70	30	0	0	0
280	30	0	0	0
640	30	0	0	0
1,100	28	0	0	0
1,800	30	0	0	0
2,500	30	0	0	0
3,500	27	0	0	0
4,600	29	0	0	0
5,800	27	0	0	0
7,000	28	0	0	0

The testing on the times of the data recovery completed in 10 seconds found that our DR Code produced almost 100% success. Actually, it should have been successful for all 30 times which implies that the increasing number of cells makes it more difficult to identify pixels and it might take it longer to decode the bigger data block. The testing on the data recovery by the three broadly use software indicated that those three software could not recover the data at all, though the simulated data loss was only 15 – 20% (less than 30%) and with 30 – 60 seconds added for the processing time. This might be that the manufacturer did not combine the functions required for the data recovery into those software.

Furthermore, regarding the testing on data recovery of the proposed DR Code, we tested it with six cases of data loss by comparing them roughly case by case and found that the DR Code could recover all data damage from those six cases and each case spent the same period of time.

VI. CONCLUSION

This study proposes the new model of the two-dimension barcode called DR Code (Data Restorable Code) which is more effective to recover the lost data. This DR Code is able to recover six cases of data loss which often found in practical implementation in both industry and logistics sectors. We have designed the model and implemented it in order to generate and read the DR Code effectively. Moreover, the test was conducted to compare the functional speed between the DR

Code and QR Code by using three broadly used QR Code readers. The result indicates that the DR Code decoding takes smaller amount of time than the QR code. The test was also conducted on the data recovery performance and found that the DR Code is able to recover all data from those six cases of the data loss.

For the next phase of this study, we aim to conduct an experiment on the curving and bending areas in order to affirm the Alignment Pattern performance compared with the Alignment Pattern of the QR code. Also, we will investigate more to point out the possible weaknesses of the proposed model and apply them to improve its performance. More importantly, we would like to conduct a study how to take advantage from the free space on the marginal areas.

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